

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 5.66659136492977
rotor delay [s]: 0.053388957398
gimbal delay [s]: 0.001752347303
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 2.1860769494558426e-14
common-scale FD null norm: 6.4746837573267136e-09

Dimensionless coordinate

log_mass_scale	0.281762438168
log_second_moment_mode_xx	-9.41561832499
log_second_moment_mode_yy	-1.97987028034
log_second_moment_mode_zz	11.5160562352
log_second_moment_mode_xy	-2.92834937811
log_second_moment_mode_xz	3.39297684371
log_second_moment_mode_yz	-1.24518736419
dimensionless_cog_displacement_x	-0.245932829253
dimensionless_cog_displacement_y	-0.390565602269
dimensionless_cog_displacement_z	0.0513619783573
log_force_effectiveness_1	0.614533885688
log_force_effectiveness_2	-0.289167055284
log_force_effectiveness_3	-1.21098708841
log_force_effectiveness_4	0.48329018995

Physical representative

mass [kg]: 3.1169574857811653
CoG body [m]: [-0.04914175 -0.07485704 0.01934993]
force effectiveness: [1.84879465 0.74888709 0.29790308 1.62140035]
inertia [kg m²]:
[128.1689601, 89.2357155, -94.35810268]
[89.2357155, 199.2581593, 63.97843407]
[-94.35810268, 63.97843407, 192.1205747]
principal moments [kg m²]: [7.85833606e-03 2.59765990e+02 2.59773846e+02]

Per-bag raw residual-wrench RMS

failure_2_single: [2.07651122e+01 2.84823899e+01 1.03184251e+01 1.59822497e+04
2.17628640e+04 1.81619869e+04]